

Framing the university's operations this way also clarifies ownership: the Student Management module, for example, plays the same role that a CRM plays in a commercial ERP — it is the system of record for who the customer (student) is and what their status is — while Employee Management mirrors a conventional HCM module used to manage staff records in any large organisation.

2. ERP Solution Design

The solution consists of specialised modules accessed via dedicated role-based portals (Admin, Lecturer, and Student), all reading and writing to the same underlying database. The goal is to clearly show how university processes connect and how data flows seamlessly between students, faculty, and administration, rather than treating each portal as an isolated application with its own private data store.

Role-based access ensures each portal only exposes the functionality relevant to that user. Administrators see system-wide configuration and analytics; lecturers see the courses, students, and grade books assigned to them; and students see only their own academic record, attendance, and results. This separation keeps the interface simple for each user type while still drawing from one unified dataset behind the scenes.

2.1 The Core Modules

Module	Function
Overview & Settings	Central command center providing live metrics, system administration, and role-based access control.
Employee Management	Manages lecturer profiles, department allocations, and specialization records.
Student Management	Handles student admissions, batch assignments, and major tracking.
Course Registration	Links students and lecturers to specific academic subjects and credit allocations.
Examination Management	Allows lecturers to input marks and automatically calculates grades for student viewing.
Attendance Management	Tracks daily class presence, instantly updating real-time attendance percentages for students.

2.2 The Part That Matters: One Shared Record

Six separate modules alone would just be isolated tools. What makes Lumina an integrated ERP is the shared Student Academic Record, which is updated by multiple modules throughout the semester lifecycle. Rather than each module owning its own private copy of student data, every module reads from and writes to the same underlying record, so an update made in one place is immediately visible everywhere else.

The lifecycle flow works as follows:

- Admin / Employees creates the student profile and registers available courses, establishing the baseline record that every later step builds on.
- Course Registration allows the student to enroll, linking them to a specific lecturer and locking in the credit allocation for that semester.
- Attendance Management updates the student's daily status, reflecting instantly on their portal so both the student and their lecturer can monitor participation in real time.

- Examination Management finalizes the loop by appending grades to the student's academic profile and automatically calculating the resulting grade.
- Dashboards aggregate this data to display live institutional statistics — total students, active courses, average attendance, and pass rates — giving administrators an at-a-glance view of institutional health.

Key Insight

A single student profile flows through every module, proving the system works as one connected enterprise environment rather than a set of independent applications stitched together after the fact.

3. Technology & Deployment

The system follows a standard, secure three-tier architecture: presentation, application, and data. Separating these concerns keeps the codebase maintainable, allows each tier to be scaled independently, and means the user interface can evolve without requiring changes to the underlying business logic or storage layer.

Tier	In Our System	Technology
Presentation	Role-based portals (Admin, Student, Lecturer)	React, HTML, CSS, JavaScript
Application	Business logic, secure routing, and REST API	Spring Boot (Java)
Data	Persistent storage of the shared Student Academic Record and all module data	Relational database (e.g., MySQL / PostgreSQL) <i>*assumed — not specified in source</i>

The presentation tier is built with React, giving each role-based portal a responsive, component-driven interface that updates in real time as data changes. The application tier, built on Spring Boot, exposes a secure REST API that enforces business rules — such as grade calculation logic and role-based permissions — before any data reaches the database. This keeps validation centralised in one place rather than duplicated across each portal.